PHIL 350: Midterm

Andrew Bacon

1. In this question you will define an operation $sort : \mathbb{N}^* \to \mathbb{N}^*$ that takes a sequence of numbers and sorts them in ascending order, yielding another sequence of numbers. You may assume that the relation $m \leq n$ as already been defined, that is, you may assume that you have defined whether one number less than or equal to another.

   (a) Consider the operation, $insert : \mathbb{N} \times \mathbb{N}^* \to \mathbb{N}^*$ where $insert(m, t)$ is the result of inserting m into the sequence t after the last number in t that m is at least as big as. So $insert(4, (1, 3, 2, 8, 1, 5)) = (1, 3, 2, \mathbf{4}, 8, 1, 5)$: the 4 has been inserted after the $1, 3$ and 2, because it is at least as big as 1, 3 and 2, and it has not been inserted any later, because 8 is bigger than it.

   Using the informal description above, calculate $insert(3, (2, 3, 4))$ and $insert(5, (4, 1, 2, 3, 1, 11))$ (3 points)

   (b) Define $insert(m, t)$ by recursion. Hint: you should hold m fixed, and define it by recursion on t. You will define it by cases, and need to appeal to the previous questions. (10 points).

   (c) Define by recursion an operation, $sort : \mathbb{N}^* \to \mathbb{N}^*$, that sorts a sequence of numbers in ascending order. So, for instance, $sort((4, 1, 9, 8, 3, 4, 3, 1, 4)) = (1, 1, 3, 3, 4, 4, 4, 8, 9)$. After you have provided your answer, you should check that it works by calculating what it does to the sequence just given: $(4, 1, 9, 8, 3, 4, 3, 1, 4)$. You may assume that your definition of *insert* behaves as it ought to. (Hint: you will only need to use your definition of *insert* above.) (10 points).

2. Prove the following facts about entailments.

   (a) $A \to B, \neg B \models \neg A$. (5 points)

   (b) If $A \models C$ and $B \models C$ then $(A \vee B) \models C$. (10 points)

   (c) If $A_1 ... A_n \models C$ and $B_1 ... B_k \models D$ then $A_1 ... A_n, B_1 ... B_k \models (C \wedge D)$. (10 points)

3. (a) Define the connective $\Box$ (see truth table on last page of exam) using $\wedge, \vee$ and $\neg$. (5 points)

   (b) Show that $\to$ and $\neg$ are together truth functionally complete. (10 points.)

(c) Show that $\uparrow$ (defined on last page) is truth functionally complete on its own. (7 points.)

(d) Show that $\wedge, \vee$ and $\to$ together are not truth functionally complete. (10 points).

4. In this question you may assume that multiplication $mult : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ has already been defined. (And you may use infix or prefix notation to represent it, as you prefer.)

   (a) Using recursion, define the exponentiation function $exp : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ where $exp(m, n) = m^n = m \times m \times .... \times m$, the result of multiplying m with itself n times. (5 points).

   (b) Consider the function f that maps the number n to the result of powering 2 to itself n times. We will simply stipulate that $f(0) = 1$. That is, $f(0) = 1$, $f(1) = 2$, $f(2) = 2^2 (= 4)$, $f(3) = 2^{2^2} (= 2^4 = 16)$, $f(4) = 2^{2^{2^2}}$ and so on. Now define it using recursion. (10 points).

   (c) Prove by induction that $exp(m, add(k, n)) = mult(exp(m, k), exp(m, n))$. (Hint: fix m and k and do induction on n.)

   You may assume any properties of multiplication and addition proved in exercises or in class. For instance, associativity of multiplication: $mult(mult(m, k), n) = mult(m, mult(k, n))$. (15 points.)

P	Q	R	$\Box(P, Q, R)$
T	T	T	T
T	T	F	F
T	F	T	F
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	T
F	F	F	F

P	Q	$P \uparrow Q$
T	T	F
T	F	T
F	T	T
F	F	T